Mathematical Analysis — Solutions & Theory

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Theory first, then a complete solution of every exercise on the five seminar sheets, in the seminar's style. Every numeric and symbolic result was checked with sympy.

Colour code used in the tables	
Result, convergent (C), exists	green
Divergent (D), wrong (in the notes)	red
Method, test, set	blue
When to use it, watch out	yellow
Does not exist (∄), inconclusive	gray

Chapter 1 — Real numbers: theory

Let A ⊆ ℝ.

Bounds

$$\mathrm{lb}(A) = \{x \in \mathbb{R} \mid x \le a,\ \forall a \in A\}, \qquad \mathrm{ub}(A) = \{x \in \mathbb{R} \mid x \ge a,\ \forall a \in A\}$$

- A is **bounded below** if lb(A) ≠ ∅, **bounded above** if ub(A) ≠ ∅, **bounded** if both.
- **min(A)** is the element of lb(A) ∩ A; **max(A)** is the element of ub(A) ∩ A. They may not exist; when they exist, they are real numbers.
- **inf(A)** is the greatest lower bound; **sup(A)** is the least upper bound. They can also be ±∞ (they live in the extended real line ℝ ∪ {±∞}).
- If A is not bounded above, sup(A) = +∞. If A is not bounded below, inf(A) = −∞. By convention, sup ∅ = −∞ and inf ∅ = +∞.
- If min(A) exists, then inf(A) = min(A). If max(A) exists, then sup(A) = max(A).

Characterisation of sup and inf

This is used in almost every proof. For A ≠ ∅ bounded above:

$$s = \sup(A) \iff \begin{cases} a \le s, & \forall a \in A \quad \text{(s is an upper bound)} \\ \forall \varepsilon > 0\ \exists a_\varepsilon \in A:\ a_\varepsilon > s - \varepsilon & \text{(nothing smaller is an upper bound)} \end{cases}$$

For inf, in the same way:

$$m = \inf(A) \iff \begin{cases} a \ge m, & \forall a \in A \\ \forall \varepsilon > 0\ \exists a_\varepsilon \in A:\ a_\varepsilon < m + \varepsilon \end{cases}$$

Completeness and density

Completeness axiom. Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Density. Between any two distinct real numbers there is a rational number and an irrational number.

Neighborhoods, interior, closure

$$V \in \mathcal{V}(x) \iff \exists \varepsilon > 0 : \ (x - \varepsilon, x + \varepsilon) \subseteq V$$

$$\operatorname{int}(A) = \{x \in \mathbb{R} \mid \exists\, V \in \mathcal{V}(x),\ V \subseteq A\}, \qquad \operatorname{cl}(A) = \{x \in \mathbb{R} \mid \forall\, V \in \mathcal{V}(x),\ V \cap A \neq \emptyset\}$$

$$\operatorname{int}(A) \subseteq A \subseteq \operatorname{cl}(A)$$

- Practical rule for unions of intervals and points: **int** removes the endpoints and the isolated points; **cl** adds the finite endpoints. An isolated point stays in cl.
- Method for Ex. 1, 2, 9: draw the set on the real line, write lb and ub (or int and cl), then read off the answer.

Chapter 1 — Solutions (Seminar 1)

Ex. 1

(a) $A = [-3, 2) \cup \{3\}$

$\text{lb}(A) = (-\infty, -3]$, $\text{ub}(A) = [3, \infty)$

$\min(A) = -3$, since $-3 \in \text{lb}(A) \cap A$

$\max(A) = 3$, since $3 \in \text{ub}(A) \cap A$

$\inf(A) = -3$, $\sup(A) = 3$

(b) $B = (-1, 1] \cup (2, \infty)$

$\text{lb}(B) = (-\infty, -1]$

$\text{ub}(B) = \varnothing$, because B is not bounded above

$\min(B) \nexists$, because $-1 \notin B$; $\max(B) \nexists$

$\inf(B) = -1$, $\sup(B) = +\infty$

(c) $C = (-5, 5) \cap \mathbb{Z} = \{-4, -3, \ldots, 3, 4\}$

$\text{lb}(C) = (-\infty, -4]$, $\text{ub}(C) = [4, \infty)$

$\min(C) = \inf(C) = -4$, $\max(C) = \sup(C) = 4$

(d) $D = \varnothing$

Every $x \in \mathbb{R}$ satisfies "$x \le a$ for all $a \in \varnothing$", since there is nothing to check. So $\text{lb}(\varnothing) = \text{ub}(\varnothing) = \mathbb{R}$.

$\min(\varnothing) \nexists$, $\max(\varnothing) \nexists$

By convention $\inf(\varnothing) = +\infty$ and $\sup(\varnothing) = -\infty$.

Ex. 2

(a) $A = \{x \in \mathbb{Q} \mid x^2 < 3\} = (-\sqrt{3}, \sqrt{3}) \cap \mathbb{Q}$

Every $x \ge \sqrt{3}$ is an upper bound.

If $y < \sqrt{3}$, by density there is $q \in \mathbb{Q}$ with $\max(y, 0) < q < \sqrt{3}$. Then $q \in A$ and $q > y$, so y is not an upper bound.

$\Longrightarrow \sup(A) = \sqrt{3}$. In the same way, $\inf(A) = -\sqrt{3}$.

$\sqrt{3} \notin \mathbb{Q} \Longrightarrow \max(A) \nexists$, $\min(A) \nexists$

(b) Complete the square:

$$x^2 - 4x + 3 = (x-2)^2 - 1 \;\ge\; -1, \quad \text{with equality at } x = 2$$

$(x - 2)^2 - 1$ takes every value $\ge -1 \Longrightarrow B = [-1, \infty)$

min(B) = inf(B) = −1

max(B) ∄, sup(B) = +∞

(c) Rewrite the general term:

$$\frac{n}{n+1} = 1 - \frac{1}{n+1} \quad \Longrightarrow \quad C = \left\{\frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \dots\right\} \subset \left[\frac{1}{2},\ 1\right)$$

The terms increase with n ⟹ min(C) = inf(C) = 1/2 (for n = 1).

All terms are < 1 ⟹ 1 is an upper bound.

For ε > 0 choose n > 1/ε. Then 1 − 1/(n+1) > 1 − ε ⟹ sup(C) = 1.

1 ∉ C ⟹ max(C) ∄

(d) D = {$2^{-k} + 3^{-m}$ | k, m ∈ ℕ}

Both terms are largest at k = m = 1:

$$\max(D) = \sup(D) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

All elements are > 0 ⟹ 0 ∈ lb(D).

For ε > 0 choose k and m with 2^{-k} < ε/2 and 3^{-m} < ε/2. That element is < ε ⟹ inf(D) = 0.

0 ∉ D ⟹ min(D) ∄

Ex. 3

(a) −S is bounded below.

S ≠ ∅ and bounded above ⟹ s = sup(S) ∈ ℝ exists, and s ≥ x for all x ∈ S.

⟹ −s ≤ −x for all x ∈ S

⟹ −s ∈ lb(−S) ⟹ −S is bounded below ⟹ m = inf(−S) exists.

(b) m = −s.

① −s ∈ lb(−S), and m is the greatest lower bound ⟹ m ≥ −s.

② m ≤ −x for all x ∈ S ⟹ −m ≥ x for all x ∈ S ⟹ −m ∈ ub(S) ⟹ −m ≥ s ⟹ m ≤ −s.

①, ② ⟹ inf(−S) = −sup(S).

Ex. 4

The inf inequality. Let a = inf f and b = inf g over D.

f(x) ≥ a and g(x) ≥ b for all x ∈ D

⟹ f(x) + g(x) ≥ a + b for all x ∈ D

⟹ a + b is a lower bound of {f(x) + g(x) | x ∈ D}

⟹ inf(f + g) ≥ a + b, because inf is the greatest lower bound.

The sup inequality. Let A = sup f and B = sup g.

$f(x) \le A$ and $g(x) \le B \Longrightarrow f(x) + g(x) \le A + B$ for all $x \in D$

$\Longrightarrow A + B$ is an upper bound $\Longrightarrow \sup(f + g) \le A + B$.

Strict examples. $D = [-1, 1]$, $f(x) = x$, $g(x) = -x$, so $f + g \equiv 0$.

$\inf f + \inf g = -1 + (-1) = -2 < 0 = \inf(f + g)$

$\sup f + \sup g = 1 + 1 = 2 > 0 = \sup(f + g)$

Ex. 5

Let $s = \sup(S)$ and $T = \{ax + b \mid x \in S\}$.

① $x \le s$ and $a > 0 \Longrightarrow ax + b \le as + b$ for all $x \in S \Longrightarrow as + b$ is an upper bound of T.

② Let $\varepsilon > 0$. By the characterisation of sup, there is $x_\varepsilon \in S$ with $x_\varepsilon > s - \varepsilon/a$. Then

$$a\,x_\varepsilon + b \;>\; a\left(s - \frac{\varepsilon}{a}\right) + b \;=\; (as + b) - \varepsilon$$

①, ② $\Longrightarrow \sup(T) = a \cdot \sup(S) + b$.

Ex. 6

- **$[-1, 1] \cup \{2\}$: yes.** $(-1, 1)$ is inside the set, so $\varepsilon = 1$ works.
- **$(-1, 1) \cap \mathbb{Q}$: no.** Every interval $(-\varepsilon, \varepsilon)$ contains irrational numbers, which are not in the set.
- **$\bigcap [-1/n, 1/n] = \{0\}$: no.** If $x \neq 0$, then $|x| > 1/n$ for n large, so only 0 is left. No interval $(-\varepsilon, \varepsilon)$ fits inside $\{0\}$.

Ex. 7

U is a neighborhood of x $\Longrightarrow$ there is $\varepsilon_1 > 0$ with $(x - \varepsilon_1, x + \varepsilon_1) \subseteq U$.

V is a neighborhood of x $\Longrightarrow$ there is $\varepsilon_2 > 0$ with $(x - \varepsilon_2, x + \varepsilon_2) \subseteq V$.

Let $\varepsilon = \min(\varepsilon_1, \varepsilon_2) > 0$. Then $(x - \varepsilon, x + \varepsilon)$ is inside both U and V.

$\Longrightarrow (x - \varepsilon, x + \varepsilon) \subseteq U \cap V \Longrightarrow U \cap V$ is a neighborhood of x.

Ex. 8

The statement needs $a \neq b$: if $a = b$, every two neighborhoods of a contain a.

Let $\varepsilon = |a - b|/2 > 0$, $U = (a - \varepsilon, a + \varepsilon)$, $V = (b - \varepsilon, b + \varepsilon)$.

If some y were in $U \cap V$, then

$$|a - b| \le |a - y| + |y - b| < \varepsilon + \varepsilon = |a - b|$$

which is impossible $\Longrightarrow U \cap V = \varnothing$.

Ex. 9

Set	int	cl
(a) $(1, 2]$	$(1, 2)$	$[1, 2]$
(b) $[-3, 2) \cup \{3\}$	$(-3, 2)$	$[-3, 2] \cup \{3\}$
(c) $(-1, 1] \cup (2, \infty)$	$(-1, 1) \cup (2, \infty)$	$[-1, 1] \cup [2, \infty)$
(d) $(-5, 5) \cap \mathbb{Z} = \{-4, \ldots, 4\}$	$\varnothing$	$\{-4, \ldots, 4\}$

For (d): no interval fits inside a finite set, so int = $\varnothing$.

Every $x \notin \mathbb{Z}$ has a neighborhood that misses the set, so cl is the set itself.

Ex. 10

$A = (0, 1) \cap \mathbb{Q}$

inf A = 0. 0 is a lower bound. For $0 < \varepsilon < 1$ there is a rational q with $0 < q < \varepsilon$ (density). Then $q \in A$ and $q < 0 + \varepsilon$, so no number above 0 is a lower bound.

sup A = 1. Same idea: for $0 < \varepsilon < 1$ there is a rational q in $(1 - \varepsilon, 1)$.

int A = $\varnothing$. Every interval $(x - \varepsilon, x + \varepsilon)$ contains an irrational number, which is not in A.

cl A = [0, 1]. If $x \in [0, 1]$, every interval around x meets $(0, 1)$ in an interval, which contains a rational $\Longrightarrow x \in \text{cl } A$.

If $x < 0$, the neighborhood $(2x, 0)$ misses A. If $x > 1$, the neighborhood $(1, 2x)$ misses A. So no other point is in cl A.

Results at a glance — Ex. 1 and Ex. 2

Set	lb	ub	min	max	inf	sup
1(a) $[-3, 2) \cup \{3\}$	$(-\infty, -3]$	$[3, \infty)$	-3	3	-3	3
1(b) $(-1, 1] \cup (2, \infty)$	$(-\infty, -1]$	$\varnothing$	$\nexists$	$\nexists$	-1	$+\infty$
1(c) $(-5, 5) \cap \mathbb{Z}$	$(-\infty, -4]$	$[4, \infty)$	-4	4	-4	4
1(d) $\varnothing$	$\mathbb{R}$	$\mathbb{R}$	$\nexists$	$\nexists$	$+\infty$	$-\infty$
2(a) $\{x \in \mathbb{Q} : x^2 < 3\}$	$(-\infty, -\sqrt{3}]$	$[\sqrt{3}, \infty)$	$\nexists$	$\nexists$	$-\sqrt{3}$	$\sqrt{3}$
2(b) $\{x^2 - 4x + 3\}$	$(-\infty, -1]$	$\varnothing$	-1	$\nexists$	-1	$+\infty$
2(c) $\{n/(n+1)\}$	$(-\infty, 1/2]$	$[1, \infty)$	$1/2$	$\nexists$	$1/2$	1
2(d) $\{2^{-k} + 3^{-m}\}$	$(-\infty, 0]$	$[5/6, \infty)$	$\nexists$	$5/6$	0	$5/6$

Chapter 2 — Sequences: theory

The limit of a sequence

(x_n) converges to $\ell \in \mathbb{R}$ if

$$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} \ \text{ such that } \ |x_n - \ell| < \varepsilon, \quad \forall n \geq N_\varepsilon$$

(x_n) tends to $+\infty$ if for every $M > 0$ there is N with $x_n > M$ for all $n \geq N$.

A convergent sequence is bounded, and its limit is unique.

Monotone convergence (Weierstrass)

A monotone and bounded sequence is convergent.

- increasing + bounded above $\Longrightarrow$ lim = sup
- decreasing + bounded below $\Longrightarrow$ lim = inf

To check monotonicity, look at the sign of $x_{n+1} - x_n$, or (for positive terms) compare x_{n+1}/x_n with 1.

Squeeze theorem

$$a_n \leq x_n \leq b_n \text{ (from some rank on)}, \quad a_n \to \ell,\ b_n \to \ell \quad \Longrightarrow \quad x_n \to \ell$$

The number e and 1^∞ limits

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n, \qquad \lim_{n\to\infty} (1 + u_n)^{\frac{1}{u_n}} = e \quad \text{whenever } u_n \to 0,\ u_n \neq 0$$

$$a_n \to 1,\ b_n \to \pm\infty \quad \Longrightarrow \quad \lim_{n\to\infty} a_n^{b_n} = e^{\lim_{n\to\infty} (a_n - 1)\, b_n}$$

For powers in general use

$$a^b = e^{b \ln a} \qquad (a > 0)$$

Useful limits

$$\sqrt[n]{n} \to 1, \qquad \sqrt[n]{a} \to 1 \ (a > 0), \qquad q^n \to 0 \ (|q| < 1), \qquad \frac{\ln n}{n} \to 0, \qquad \frac{n^k}{a^n} \to 0 \ (a > 1), \qquad \frac{a^n}{n!} \to 0$$

Ratio test for sequences

Let $x_n > 0$ and

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \ell. \qquad \ell < 1 \ \Rightarrow \ \lim x_n = 0; \qquad \ell > 1 \ \Rightarrow \ \lim x_n = \infty$$

Ratio ⟹ root (Ex. 7):

$$\frac{x_{n+1}}{x_n} \to \ell \quad \Longrightarrow \quad \sqrt[n]{x_n} \to \ell$$

Stolz–Cesàro

Let (a_n), (b_n) satisfy (i) $a_n \to 0$, $b_n \to 0$, (b_n) strictly decreasing; or (ii) $b_n \to \infty$, (b_n) strictly increasing. Then

$$\lim_{n\to\infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \ell \quad \Longrightarrow \quad \lim_{n\to\infty} \frac{a_n}{b_n} = \ell$$

Use it when the numerator is a sum of n terms ($1 + 1/2 + \ldots + 1/n$, $1^p + \ldots + n^p$) or an average.

Chapter 2 — Solutions (Seminar 2)

Ex. 1

Let ε > 0. We look for Nε ∈ ℕ such that |1/√n − 0| < ε for all n ≥ Nε.

$$\frac{1}{\sqrt{n}} < \varepsilon \iff \sqrt{n} > \frac{1}{\varepsilon} \iff n > \frac{1}{\varepsilon^2}$$

Choose

$$N_\varepsilon = \left[\frac{1}{\varepsilon^2}\right] + 1$$

Then n ≥ Nε > 1/ε² ⟹ 1/√n < ε for all n ≥ Nε.

Since ε > 0 was arbitrary ⟹

$$\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$$

Ex. 2

(a) Multiply by the conjugate:

$$x_n = \sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

Bounded. √(n+1) + √n ≥ √2 + 1 for all n ≥ 1, so

$$0 < x_n \le \frac{1}{\sqrt{2}+1} = \sqrt{2} - 1 < 1 \qquad (1)$$

Monotone. The denominator increases with n:

$$\sqrt{n+1} + \sqrt{n} < \sqrt{n+2} + \sqrt{n+1} \implies x_n > x_{n+1} \quad \text{(decreasing)} \qquad (2)$$

①, ② ⟹ (Weierstrass) convergent. The denominator → ∞, so lim x_n = 0.

(b) Split each term:

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

The sum telescopes:

$$x_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1}$$

Monotone.

$$x_{n+1} - x_n = \frac{1}{(n+1)(n+2)} > 0 \implies \text{increasing} \qquad (1)$$

Bounded. $1/2 = x_1 \le x_n < 1$ ②

①, ② $\Longrightarrow$ convergent, and $\lim x_n = 1$.

(c) $x_n = 2^n/n! > 0$. Look at the ratio:

$$\frac{x_{n+1}}{x_n} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1}$$

For $n = 1$ the ratio is 1, so $x_2 = x_1 = 2$.

For $n \ge 2$ the ratio is < 1, so $x_{n+1} < x_n$.

$\Longrightarrow$ (x_n) is decreasing (non-increasing) $\Longrightarrow$ $0 < x_n \le x_1 = 2$ $\Longrightarrow$ bounded $\Longrightarrow$ convergent.

Ratio test for sequences: $2/(n+1) \to 0 < 1 \Longrightarrow \lim x_n = 0$.

Ex. 3

(a) Conjugate, then divide by $\sqrt{n}$:

$$\sqrt{n}\left(\sqrt{n+1} - \sqrt{n}\right) = \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \longrightarrow \frac{1}{2}$$

(b) Squeeze:

$$\frac{2^n - 1}{3^n} \le \frac{2^n + (-1)^n}{3^n} \le \frac{2^n + 1}{3^n}$$

Both bounds are $(2/3)^n \pm (1/3)^n \to 0$, so the limit is **0**.

(c) Write the root as a power of e:

$$\sqrt[n]{n} = n^{\frac{1}{n}} = e^{\frac{\ln n}{n}} \longrightarrow e^0 = 1, \qquad \text{since } \frac{\ln n}{n} \to 0$$

(d) Factor out n^2:

$$\frac{(an+1)^2}{4n^2 - 2n + 1} = \frac{n^2\left(a + \frac{1}{n}\right)^2}{n^2\left(4 - \frac{2}{n} + \frac{1}{n^2}\right)} \longrightarrow \frac{a^2}{4}$$

For $a = 0$ the numerator is 1 and the limit is $0 = a^2/4$, so the answer **$a^2/4$** holds for every $a \in \mathbb{R}$.

(e) First the example from the seminar:

$$(3^n + 4^n)^{\frac{1}{n}} = 4\left(\left(\frac{3}{4}\right)^n + 1\right)^{\frac{1}{n}} \longrightarrow 4 \cdot 1 = 4$$

In general, let $M = \max\{a_1, \ldots, a_k\}$. Then

$$M^n \le a_1^n + \cdots + a_k^n \le k\,M^n \quad \Longrightarrow \quad M \le x_n \le \sqrt[n]{k} \cdot M$$

$\sqrt[n]{k} \to 1$, so by squeeze $\lim x_n =$ **$\max\{a_1, \ldots, a_k\}$**.

Ex. 4

Increasing. Apply the AM–GM inequality to the n + 1 numbers 1, 1 + 1/n, …, 1 + 1/n (n copies):

$$\underbrace{\left(1\cdot\left(1+\frac{1}{n}\right)^{n}\right)^{\frac{1}{n+1}}}_{\text{geometric mean}} \le \underbrace{\frac{1+n\left(1+\frac{1}{n}\right)}{n+1}}_{\text{arithmetic mean}} = \frac{n+2}{n+1} = 1+\frac{1}{n+1}$$

Raise both sides to the power n + 1:

$$\left(1+\frac{1}{n}\right)^{n} \le \left(1+\frac{1}{n+1}\right)^{n+1} \quad\Longleftrightarrow\quad e_n \le e_{n+1}$$

Bounded. $e_n \ge e_1 = 2$. By the binomial theorem:

$$e_n = \sum_{k=0}^{n} C_n^k \frac{1}{n^k} = \sum_{k=0}^{n} \frac{n(n-1)\cdots(n-k+1)}{n^k}\cdot\frac{1}{k!} \le \sum_{k=0}^{n}\frac{1}{k!}$$

Using $k! \ge 2^{k-1}$:

$$\sum_{k=0}^{n}\frac{1}{k!} \le 1+1+\frac{1}{2}+\frac{1}{2^2}+\cdots+\frac{1}{2^{n-1}} < 1+2=3$$

So $2 \le e_n < 3 \Longrightarrow$ (Weierstrass) (e_n) converges; its limit e lies in [2, 3].

Ex. 5

(a) It is a 1^∞ limit. Write (2n+1)/(2n−1) = 1 + 2/(2n−1):

$$x_n = \left[\left(1+\frac{2}{2n-1}\right)^{\frac{2n-1}{2}}\right]^{\frac{2n}{2n-1}} \longrightarrow e^{\lim\frac{2n}{2n-1}} = e^1 = e$$

(b) Combine the logarithms:

$$n\big(\ln(n+2)-\ln(n+1)\big) = \ln\left(1+\frac{1}{n+1}\right)^{n} = \ln\left[\left(1+\frac{1}{n+1}\right)^{n+1}\right]^{\frac{n}{n+1}} \longrightarrow \ln e^1 = 1$$

Ex. 6

(a) Let $y_n = x_1 + \ldots + x_n$ and $z_n = n$. (z_n) is strictly increasing and $z_n \to \infty$, and $a_n = y_n/z_n$.

$$\frac{y_{n+1}-y_n}{z_{n+1}-z_n} = \frac{x_{n+1}}{1} \longrightarrow \ell = \lim_{n\to\infty} x_n$$

$\Longrightarrow$ (Stolz–Cesàro) $a_n \to \ell$: **the averages converge to the same limit.**

The converse is false. $x_n = (-1)^n$ does not converge, but its averages do:

$$|a_n| = \frac{|(-1)+1+(-1)+\cdots+(-1)^n|}{n} \le \frac{1}{n} \longrightarrow 0$$

(b) Let $H_n = 1 + 1/2 + \ldots + 1/n$.

Compared with n (Stolz–Cesàro (ii), $b_n = n$):

$$\frac{H_{n+1} - H_n}{(n+1) - n} = \frac{1}{n+1} \longrightarrow 0 \quad \Longrightarrow \quad \frac{H_n}{n} \to 0$$

Compared with ln n ($b_n = \ln n$ is strictly increasing, $\to \infty$):

$$\frac{H_{n+1} - H_n}{\ln(n+1) - \ln n} = \frac{\frac{1}{n+1}}{\ln\left(1 + \frac{1}{n}\right)} = \frac{1}{\ln\left(1 + \frac{1}{n}\right)^{n+1}} \longrightarrow \frac{1}{\ln e} = 1 \quad \Longrightarrow \quad \frac{H_n}{\ln n} \to 1$$

So H_n grows like ln n — much slower than n.

Ex. 7

Assume $x_n > 0$ and $x_{n+1}/x_n \to \ell \in (0, \infty)$. Write the root as a power of e:

$$\sqrt[n]{x_n} = x_n^{\frac{1}{n}} = e^{\frac{\ln x_n}{n}}$$

Stolz–Cesàro (ii) with $a_n = \ln x_n$ and $b_n = n$:

$$\frac{\ln x_{n+1} - \ln x_n}{(n+1) - n} = \ln \frac{x_{n+1}}{x_n} \longrightarrow \ln \ell \quad \Longrightarrow \quad \frac{\ln x_n}{n} \to \ln \ell \quad \Longrightarrow \quad \sqrt[n]{x_n} \to e^{\ln \ell} = \ell$$

If $\ell = 0$, the same argument gives $(\ln x_n)/n \to -\infty$ (Stolz–Cesàro also holds for infinite limits), so $\sqrt[n]{x_n} \to 0 = \ell$.

Ex. 8

(a) Write the term as an n-th root and use Ex. 7:

$$x_n = \frac{n}{\sqrt[n]{n!}} = \sqrt[n]{\frac{n^n}{n!}}, \qquad y_n = \frac{n^n}{n!}$$

$$\frac{y_{n+1}}{y_n} = \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \frac{(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n \longrightarrow e \quad \Longrightarrow \quad \lim_{n\to\infty} x_n = e$$

(b) Stolz–Cesàro (ii) with $a_n = 1^p + \ldots + n^p$ and $b_n = n^{(p+1)}$ (strictly increasing, $\to \infty$). Expand the denominator with the binomial theorem:

$$\frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \frac{(n+1)^p}{(n+1)^{p+1} - n^{p+1}} = \frac{(n+1)^p}{(p+1)\,n^p + C_{p+1}^2\, n^{p-1} + \cdots + 1} \longrightarrow \frac{1}{p+1}$$

(divide numerator and denominator by n^p) $\Longrightarrow$

$$\lim_{n\to\infty} \frac{1^p + 2^p + \cdots + n^p}{n^{p+1}} = \frac{1}{p+1}$$

Results at a glance — Seminar 2

Exercise	Method	Result
1	ε-definition, $N = [1/\varepsilon^2] + 1$	lim = 0

Exercise	Method	Result
2(a) $\sqrt{(n+1)} - \sqrt{n}$	conjugate; Weierstrass	decreasing, bounded, lim = 0
2(b) $\Sigma\ 1/(k(k+1))$	telescoping; Weierstrass	increasing, bounded, lim = 1
2(c) $2^n/n!$	ratio $2/(n+1)$	non-increasing, bounded, lim = 0
3(a) $\sqrt{n}(\sqrt{(n+1)} - \sqrt{n})$	conjugate	1/2
3(b) $(2^n + (-1)^n)/3^n$	squeeze	0
3(c) $\sqrt[n]{n}$	e^((ln n)/n)	1
3(d) $(an+1)^2/(4n^2 - 2n + 1)$	highest power of n	$a^2/4$
3(e) $(a_1^n + \ldots + a_k^n)$^(1/n)	factor out the largest, squeeze	max a_i
4 $e_n = (1 + 1/n)^n$	AM–GM; binomial theorem	increasing, $2 \le e_n < 3$
5(a) $((2n+1)/(2n-1))^n$	number e	e
5(b) $n(\ln(n+2) - \ln(n+1))$	number e	1
6(a) averages	Stolz–Cesàro	same limit ℓ
6(b) H_n/n, $H_n/\ln n$	Stolz–Cesàro	0 and 1
8(a) $n/\sqrt[n]{(n!)}$	ratio $\Longrightarrow$ root	e
8(b) $(1^p + \ldots + n^p)$/n^(p+1)	Stolz–Cesàro	1/(p+1)

Chapter 3 — Series: theory

The series $\Sigma\, x_n$ **converges (C)** if its partial sums $S_n = x_1 + \ldots + x_n$ converge; the sum is $\lim S_n$. Otherwise it **diverges (D)**.

Necessary condition

$$\sum x_n \text{ C} \quad \Longrightarrow \quad x_n \to 0 \qquad \Longleftrightarrow \qquad x_n \not\to 0 \quad \Longrightarrow \quad \sum x_n \text{ D}$$

The converse is false: $1/n \to 0$, but the harmonic series $\Sigma\, 1/n$ diverges.

Reference series

$$\sum_{n\geq 0} q^n = \frac{1}{1-q} \;\; (|q|<1), \qquad \sum_{n\geq 1} \frac{1}{n^p} \text{ is C} \iff p>1, \qquad \sum_{n\geq 0} \frac{x^n}{n!} = e^x$$

The geometric series diverges for $|q| \geq 1$. The harmonic series ($p = 1$) diverges.

Telescoping

$$x_n = y_n - y_{n+1} \quad \Longrightarrow \quad S_n = y_1 - y_{n+1} \quad \Longrightarrow \quad \sum_{n\geq 1} x_n = y_1 - \lim_{n\to\infty} y_n$$

Split the general term with partial fractions first. A convergent series may not be split into two divergent series.

Convergence tests for series with positive terms ($x_n > 0$)

T_1 Comparison test. If $x_n \leq y_n$ from some rank n_0 on:

$$\sum y_n \text{ C} \implies \sum x_n \text{ C}; \qquad \sum x_n \text{ D} \implies \sum y_n \text{ D}$$

T_2 Limit comparison test.

$$\ell = \lim_{n\to\infty} \frac{x_n}{y_n}: \qquad \ell \in (0,\infty) \Rightarrow \text{same nature}; \qquad \ell = 0,\ \textstyle\sum y_n \text{ C} \Rightarrow \sum x_n \text{ C}; \qquad \ell = \infty,\ \textstyle\sum y_n \text{ D} \Rightarrow \sum x_n \text{ D}$$

T_3 Root test, T_4 Ratio test, T_5 Raabe–Duhamel test:

$$\ell = \lim_{n\to\infty} \sqrt[n]{x_n}, \qquad \ell = \lim_{n\to\infty} \frac{x_{n+1}}{x_n}, \qquad R = \lim_{n\to\infty} n\left(\frac{x_n}{x_{n+1}} - 1\right)$$

T_6 Cauchy condensation test. If (x_n) is decreasing and $x_n > 0$:

$$\sum_{n\geq 1} x_n \quad \text{and} \quad \sum_{n\geq 0} 2^n\, x_{2^n} \quad \text{have the same nature.}$$

Which test, when

Test	Try it when x_n has…	C if	D if	Inconclusive
T_1 / T_2 Comparison	n to a constant power, ln n, sin, ln(1 + 1/n)	compared with a C series	compared with a D series	—
T_3 Root	$(\dots)^n$ or $(\dots)^{(n^2)}$	$\ell < 1$	$\ell > 1$	$\ell = 1$
T_4 Ratio	n!, a^n, products $1\cdot3\cdot\dots\cdot(2n-1)$	$\ell < 1$	$\ell > 1$	$\ell = 1 \to$ try T_5
T_5 Raabe–Duhamel	the ratio test gave $\ell = 1$	$R > 1$	$R < 1$	$R = 1$
T_6 Condensation	ln n next to n in the denominator	condensed series C	condensed series D	—

Inequalities often used: $\ln(1 + x) < x$ for $x > 0$; $\ln x < x$ for $x > 0$; $e^x \geq 1 + x$; $\sin x \leq x$ for $x \geq 0$.

Chapter 3 — Solutions (Seminar 3)

Ex. 1 — sums

(a) Geometric series with q = 1/3:

$$\sum_{n\ge 1}\frac{2}{3^n} = 2\sum_{n\ge 1}\left(\frac{1}{3}\right)^n = 2\cdot\frac{\frac{1}{3}}{1-\frac{1}{3}} = 2\cdot\frac{1}{2} = 1$$

(b) Split the numerator. Both parts give convergent series, so the split is allowed:

$$\frac{2n+1}{n!} = \frac{2n}{n!} + \frac{1}{n!} = \frac{2}{(n-1)!} + \frac{1}{n!}$$

$$\sum_{n\ge 1}\frac{2n+1}{n!} = 2\sum_{k\ge 0}\frac{1}{k!} + \left(\sum_{k\ge 0}\frac{1}{k!} - 1\right) = 2e + (e-1) = 3e - 1$$

(The series of e is Σ 1/k! from k = 0; the second sum misses the term k = 0, which is 1.)

(c) Partial fractions:

$$\frac{1}{4n^2-1} = \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$$

The partial sum telescopes:

$$S_n = \frac{1}{2}\left(1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \cdots + \frac{1}{2n-1} - \frac{1}{2n+1}\right) = \frac{1}{2}\left(1 - \frac{1}{2n+1}\right) \longrightarrow \frac{1}{2}$$

(d) Telescoping in two steps:

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

Check: the bracket equals ((n+2) − n)/(n(n+1)(n+2)) = 2/(n(n+1)(n+2)).

$$S_n = \frac{1}{2}\left(\frac{1}{1\cdot 2} - \frac{1}{(n+1)(n+2)}\right) \longrightarrow \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

(e) Differentiate the geometric series (Seminar 4, Ex. 2), then put x = 1/2:

$$\sum_{n\ge 1} n\,x^n = \frac{x}{(1-x)^2} \quad\Longrightarrow\quad \sum_{n\ge 1}\frac{n}{2^n} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

(f) Compare with the harmonic series:

$$\frac{1}{\sqrt[3]{n}} = \frac{1}{n^{1/3}} \ge \frac{1}{n} \quad\text{and}\quad \sum\frac{1}{n}\ \mathrm{D} \quad\Longrightarrow\quad \sum_{n\ge 1}\frac{1}{\sqrt[3]{n}}\ \mathrm{D}$$

The partial sums tend to +∞, so the “sum” is **+∞**.

Ex. 2 — convergence

(a) ln n < n for n ≥ 2, so

$$\frac{1}{\ln n} > \frac{1}{n} \quad \text{and} \quad \sum \frac{1}{n}\ \mathrm{D} \quad \xRightarrow{T_1} \quad \sum_{n\geq 2} \frac{1}{\ln n}\ \mathrm{D}$$

(b) Compare with a p-series, p = 3/2:

$$\frac{1}{n\sqrt{n+1}} < \frac{1}{n\sqrt{n}} = \frac{1}{n^{3/2}} \quad \text{and} \quad \sum \frac{1}{n^{3/2}}\ \mathrm{C} \quad \xRightarrow{T_1} \quad \mathrm{C}$$

(c) Use ln(1 + x) < x for x > 0:

$$\frac{\ln\left(1+\frac{1}{n}\right)}{n} < \frac{1}{n^2} \quad \text{and} \quad \sum \frac{1}{n^2}\ \mathrm{C} \quad \xRightarrow{T_1} \quad \mathrm{C}$$

(d) Ratio test:

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)!}{(n+2)^{n+1}} \cdot \frac{(n+1)^n}{n!} = \left(\frac{n+1}{n+2}\right)^{n+1} = \frac{1}{\left(1+\frac{1}{n+1}\right)^{n+1}} \longrightarrow \frac{1}{e} < 1 \quad \Longrightarrow \quad \mathrm{C}$$

(e) Root test:

$$\sqrt[n]{x_n} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \longrightarrow \frac{1}{e} < 1 \quad \Longrightarrow \quad \mathrm{C}$$

(f) x_n = 1/(n ln n) is positive and decreasing for n ≥ 2, so use condensation (T_6):

$$2^n\, x_{2^n} = 2^n \cdot \frac{1}{2^n \ln 2^n} = \frac{1}{n \ln 2} \qquad \text{and} \qquad \sum \frac{1}{n \ln 2} = \frac{1}{\ln 2} \sum \frac{1}{n}\ \mathrm{D}$$

⟹ Σ 1/(n ln n) is **D**.

Ex. 3

(a) Ratio test first:

$$\frac{x_{n+1}}{x_n} = \frac{2n+1}{2n+2} \longrightarrow 1 \quad \Longrightarrow \quad \text{inconclusive}$$

Then Raabe–Duhamel:

$$R = \lim_{n\to\infty} n\left(\frac{2n+2}{2n+1} - 1\right) = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1 \quad \Longrightarrow \quad \mathrm{D}$$

(b) Rewrite the term as a power of n:

$$x_n = a^{\ln n} = e^{\ln a \cdot \ln n} = \left(e^{\ln n}\right)^{\ln a} = n^{\ln a} = \frac{1}{n^{-\ln a}}$$

It is a p-series with p = −ln a:

- p > 1 ⟺ −ln a > 1 ⟺ a < 1/e ⟹ **C**
- p ≤ 1 ⟺ a ≥ 1/e ⟹ **D**

(c) Ratio test:

$$\frac{x_{n+1}}{x_n} = \frac{a^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{a^n\, n!} = a\Big(\frac{n}{n+1}\Big)^n = \frac{a}{\big(1+\frac{1}{n}\big)^n} \longrightarrow \frac{a}{e}$$

- a < e $\Longrightarrow$ a/e < 1 $\Longrightarrow$ **C**
- a > e $\Longrightarrow$ a/e > 1 $\Longrightarrow$ **D**
- a = e: the ratio test is inconclusive. But $(1 + 1/n)^n$ increases strictly to e, so

$$\frac{x_{n+1}}{x_n} = \frac{e}{\big(1+\frac{1}{n}\big)^n} > 1 \quad\Longrightarrow\quad x_n > x_{n-1} > \cdots > x_1 = e \quad\Longrightarrow\quad x_n \not\to 0 \quad\Longrightarrow\quad \text{D}$$

Conclusion: C for a ∈ (0, e), D for a ≥ e.

Results at a glance — Seminar 3

Exercise	Method	Result
1(a) Σ $2/3^n$	geometric series	1
1(b) Σ (2n+1)/n!	split + Σ 1/k! = e	3e − 1
1(c) Σ $1/(4n^2 - 1)$	partial fractions, telescoping	1/2
1(d) Σ 1/(n(n+1)(n+2))	telescoping	1/4
1(e) Σ $n/2^n$	differentiated geometric series	2
1(f) Σ $1/\sqrt[3]{n}$	comparison with Σ 1/n	+∞ (D)
2(a) Σ 1/ln n	comparison with Σ 1/n	D
2(b) Σ 1/(n√(n+1))	comparison with Σ 1/n^(3/2)	C
2(c) Σ ln(1 + 1/n)/n	comparison with Σ $1/n^2$	C
2(d) Σ $n!/(n+1)^n$	ratio test, ℓ = 1/e	C
2(e) Σ (n/(n+1))^(n^2)	root test, ℓ = 1/e	C
2(f) Σ 1/(n ln n)	condensation	D
3(a) Σ (1·3·…·(2n−1))/(2·4·…·2n)	ratio = 1, then Raabe R = 1/2	D
3(b) Σ a^(ln n)	p-series, p = −ln a	C if a < 1/e; D if a ≥ 1/e
3(c) Σ $a^n\, n!/n^n$	ratio test, ℓ = a/e	C if a < e; D if a ≥ e

Chapter 4 — Alternating series and power series: theory

For the exam: first check whether the series has positive terms or alternating terms. If it is alternating, try the Leibniz test.

Absolute convergence

$\Sigma\, x_n$ is **absolutely convergent (A.C.)** if $\Sigma\, |x_n|$ converges.

$$\sum |x_n| \text{ C} \quad \Longrightarrow \quad \sum x_n \text{ C}$$

A series that converges but is not A.C. is **conditionally convergent**. To test A.C., apply the positive-term tests to $|x_n|$.

Leibniz test

$$(a_n) \text{ decreasing}, \quad a_n \to 0 \quad \Longrightarrow \quad \sum (-1)^n a_n \text{ C}$$

Power series

For a sequence (a_n) and a centre $c \in \mathbb{R}$:

$$\sum_{n \geq 0} a_n \, (x - c)^n$$

There is a unique $R \in [0, \infty]$, the **radius of convergence**, such that the series converges absolutely for $|x - c| < R$ and diverges for $|x - c| > R$.

Computing R

If one of the limits below exists in $[0, \infty]$, then

$$L = \lim_{n \to \infty} \sqrt[n]{|a_n|} \quad \text{or} \quad L = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|, \qquad R = \begin{cases} \dfrac{1}{L}, & L \in (0, \infty) \\ \infty, & L = 0 \\ 0, & L = \infty \end{cases}$$

The convergence set

$$(c - R,\ c + R) \subseteq \mathcal{C} \subseteq [c - R,\ c + R]$$

The endpoints $x = c - R$ and $x = c + R$ must be tested separately: substitute them and use the tests for numerical series.

Term-by-term differentiation and integration

Inside (c − R, c + R) a power series can be differentiated and integrated term by term; the new series has the same R.

$$\sum_{n\ge 0} t^n = \frac{1}{1-t} \quad (|t|<1) \qquad \Longrightarrow \qquad \sum_{n\ge 1} n\,t^{n-1} = \frac{1}{(1-t)^2}, \qquad \sum_{n\ge 0} \frac{x^{n+1}}{n+1} = -\ln(1-x)$$

Abel’s theorem

If a power series converges at an endpoint, its sum is continuous there (from inside the interval). So an identity proved on (c − R, c + R) extends to every endpoint where the series converges.

Chapter 4 — Solutions (Seminar 4)

Ex. 1

(a) Let

$$a_n = \frac{1}{\sqrt{n(n+1)}}$$

Convergence. n(n+1) increases, so (a_n) is decreasing, and $a_n \to 0 \implies$ (Leibniz) **C**.

Absolute convergence. Compare $|x_n|$ with 1/(n+1):

$$|x_n| = \frac{1}{\sqrt{n(n+1)}} \geq \frac{1}{\sqrt{(n+1)^2}} = \frac{1}{n+1} \quad \text{and} \quad \sum \frac{1}{n+1} \text{ D} \quad \implies \quad \sum |x_n| \text{ D}$$

$\implies$ **not A.C.** The series is conditionally convergent.

(b) *Convergence.* $1/n \in (0, 1] \subset (0, \pi/2)$, where sin is increasing and positive. So sin(1/n) is decreasing, positive and $\to 0 \implies$ (Leibniz) **C**.

Absolute convergence. Limit comparison with the harmonic series:

$$\lim_{n\to\infty} \frac{\sin\frac{1}{n}}{\frac{1}{n}} = 1 \in (0,\infty) \quad \overset{T_2}{\implies} \quad \sum \sin\frac{1}{n} \text{ has the nature of } \sum \frac{1}{n} \text{ (D)}$$

$\implies$ **not A.C.** The series is conditionally convergent.

(c) The signs of sin n are irregular, so the series is not alternating. Test A.C. directly:

$$\left|\frac{\sin n}{n^2}\right| \leq \frac{1}{n^2} \quad \text{and} \quad \sum \frac{1}{n^2} \text{ C} \quad \overset{T_1}{\implies} \quad \text{A.C.} \quad \implies \quad \text{C}$$

Ex. 2

Start from the geometric series, which is A.C. for $|x| < 1$:

$$S(x) = \sum_{n\geq 0} x^n = \frac{1}{1-x}$$

Differentiate once, then multiply by x:

$$S'(x) = \sum_{n\geq 1} n\,x^{n-1} = \frac{1}{(1-x)^2} \;\Big|\cdot x \quad \implies \quad \sum_{n\geq 1} n\,x^n = \frac{x}{(1-x)^2}$$

Differentiate again, then multiply by x^2:

$$S''(x) = \sum_{n\geq 2} n(n-1)\,x^{n-2} = \frac{2}{(1-x)^3} \;\Big|\cdot x^2 \quad \implies \quad \sum_{n\geq 2} n(n-1)\,x^n = \frac{2x^2}{(1-x)^3}$$

Ex. 3

Integrate the geometric series from 0 to x ($|x| < 1$):

$$\sum_{n\geq 0}\int_0^x t^n\,dt = \int_0^x \frac{dt}{1-t} \quad\Longleftrightarrow\quad \sum_{n\geq 0}\frac{x^{n+1}}{n+1} = -\ln(1-x) \quad\Longleftrightarrow\quad \sum_{n\geq 1}\frac{x^n}{n} = -\ln(1-x)$$

Replace x by $-x$ (still $|-x| < 1$) and multiply by -1:

$$\sum_{n\geq 1}\frac{(-1)^n x^n}{n} = -\ln(1+x) \quad\Big|\cdot(-1) \quad\Longrightarrow\quad \sum_{n\geq 1}(-1)^{n+1}\frac{x^n}{n} = \ln(1+x)$$

Ex. 4

In the geometric series replace t by $-t$, then t by t^2 ($|t| < 1 \Longrightarrow t^2 \in [0, 1)$):

$$\sum_{n\geq 0}(-1)^n t^{2n} = \frac{1}{1+t^2}$$

Integrate from 0 to x, for $|x| < 1$:

$$\sum_{n\geq 0}(-1)^n\frac{x^{2n+1}}{2n+1} = \int_0^x \frac{dt}{1+t^2} = \arctan x, \qquad x\in(-1,1)$$

x = 1. The series is $\Sigma\,(-1)^n/(2n+1)$. The sequence $1/(2n+1)$ is decreasing and $\to 0$, so (Leibniz) it converges. By Abel's theorem its sum is lim arctan x as $x \to 1^-$:

$$1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots = \sum_{n\geq 0}\frac{(-1)^n}{2n+1} = \arctan 1 = \frac{\pi}{4}$$

x = −1. The series is minus the previous one, so it converges to $-\pi/4 = \arctan(-1)$. Hence the identity holds on $[-1, 1]$.

Ex. 5

(a) c = 0 and $a_n = 1/2^n$:

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{2^n}{2^{n+1}} = \frac{1}{2} \quad\Longrightarrow\quad R = 2, \qquad (-2,\,2)\subseteq\mathcal{C}$$

x = 2: $\Sigma\,1$, D. x = −2: $\Sigma\,(-1)^n$, D (the terms do not tend to 0). $\Longrightarrow$ convergence set **(−2, 2)**

(b) c = 1 and $a_n = 2^n$:

$$L = \lim_{n\to\infty}\frac{2^{n+1}}{2^n} = 2 \quad\Longrightarrow\quad R = \frac{1}{2}, \qquad \left(\frac{1}{2},\,\frac{3}{2}\right)\subseteq\mathcal{C}$$

x = 1/2: $x - 1 = -1/2$, so the series is $\Sigma\,2^n(-1/2)^n = \Sigma\,(-1)^n$, D. x = 3/2: $\Sigma\,1$, D. $\Longrightarrow$ convergence set **(1/2, 3/2)**

(c) c = 2 and $a_n = 1/((n+1)3^n)$:

$$L = \lim_{n\to\infty} \frac{(n+1)\,3^n}{(n+2)\,3^{n+1}} = \lim_{n\to\infty} \frac{n+1}{3(n+2)} = \frac{1}{3} \quad\Longrightarrow\quad R = 3, \qquad (-1,\,5) \subseteq \mathcal{C}$$

x = −1: the series is Σ $(-3)^n/((n+1)3^n)$ = Σ $(-1)^n/(n+1)$, C by Leibniz.

x = 5: the series is Σ 1/(n+1), D. ⟹ convergence set **[−1, 5)**

(d) c = 0 and $a_n = n!$:

$$L = \lim_{n\to\infty} \frac{(n+1)!}{n!} = \lim_{n\to\infty} (n+1) = \infty \quad\Longrightarrow\quad R = 0$$

The series converges only at its centre ⟹ convergence set **{0}**

(e) c = 1 and $a_n = 1/n^p$, p > 0:

$$L = \lim_{n\to\infty} \frac{n^p}{(n+1)^p} = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^p = 1 \quad\Longrightarrow\quad R = 1, \qquad (0,\,2) \subseteq \mathcal{C}$$

x = 0: Σ $(-1)^n/n^p$. The sequence $1/n^p$ is decreasing and → 0, so it converges by Leibniz, for every p > 0.

x = 2: Σ $1/n^p$, a p-series: C if p > 1, D if p ≤ 1.

$$\mathcal{C} = \begin{cases} [0,\,2), & p \in (0,\,1] \\ [0,\,2], & p > 1 \end{cases}$$

(f) c = 0 and $a_n = (-1)^n/\sqrt{n}$:

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n+1}} = 1 \quad\Longrightarrow\quad R = 1, \qquad (-1,\,1) \subseteq \mathcal{C}$$

x = 1: Σ $(-1)^n/\sqrt{n}$, C by Leibniz ($1/\sqrt{n}$ decreases to 0).

x = −1: $(-1)^n(-1)^n/\sqrt{n} = 1/\sqrt{n}$, so the series is Σ $1/\sqrt{n}$, a p-series with p = 1/2 ≤ 1, D. ⟹ convergence set **(−1, 1]**

Results at a glance — Ex. 5

Series	c	R	Left endpoint	Right endpoint	Convergence set
(a) Σ $x^n/2^n$	0	2	x = −2: D	x = 2: D	(−2, 2)
(b) Σ $2^n(x-1)^n$	1	1/2	x = 1/2: D	x = 3/2: D	(1/2, 3/2)
(c) Σ $(x-2)^n/((n+1)3^n)$	2	3	x = −1: C (Leibniz)	x = 5: D	[−1, 5)
(d) Σ $n!\,x^n$	0	0	—	—	{0}
(e) Σ $(x-1)^n/n^p$	1	1	x = 0: C (Leibniz)	x = 2: C if p > 1, D if p ≤ 1	[0, 2) or [0, 2]
(f) Σ $(-1)^n x^n/\sqrt{n}$	0	1	x = −1: D	x = 1: C (Leibniz)	(−1, 1]

Chapter 5 — Limits, continuity, derivatives: theory

Accumulation points

x ∈ ℝ is an **accumulation point** of A if every neighborhood of x contains a point of A different from x. Equivalently:

$$\exists\,(a_n) \subseteq A \setminus \{x\} \quad \text{with} \quad a_n \to x$$

The set of accumulation points is written A′. Isolated points of A are not in A′.

Limit of a function (sequential criterion)

Let $x_0 \in A'$. Then

$$\lim_{x \to x_0} f(x) = \ell \iff f(x_n) \to \ell \quad \text{for every } (x_n) \subseteq A \setminus \{x_0\} \quad \text{with} \quad x_n \to x_0$$

To show that a limit does not exist, find two sequences that give different limits.

Continuity

f is continuous at $x_0 \in A$ if

$$\lim_{x \to x_0} f(x) = f(x_0)$$

Sums, products, quotients and compositions of continuous functions are continuous; |f| is continuous when f is.

Intermediate value theorem (Bolzano). If f : [a, b] → ℝ is continuous and f(a)·f(b) ≤ 0, then f(c) = 0 for some c ∈ [a, b].

Derivative

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Differentiable at $x_0 \Longrightarrow$ continuous at x_0. The converse is false.

Fundamental limits

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \qquad \lim_{x \to 0} \frac{\ln(1+x)}{x} = 1, \qquad \lim_{x \to 0} (1+x)^{\frac{1}{x}} = e$$

Formulas and tools used

$$\sin a - \sin b = 2 \cos \frac{a+b}{2} \sin \frac{a-b}{2}, \qquad \cos a - \cos b = -2 \sin \frac{a+b}{2} \sin \frac{a-b}{2}, \qquad 1 - \cos x = 2 \sin^2 \frac{x}{2}$$

$$u \to 1,\ v \to \pm\infty \implies \lim u^v = e^{\lim (u-1)v}; \qquad u^v = e^{v \ln u}$$

L'Hôpital's rule. For 0/0 or ∞/∞ forms, lim f/g = lim f′/g′ when the right-hand limit exists.

Taylor expansions at 0:

$$e^x = 1 + x + \frac{x^2}{2} + o(x^2), \quad \ln(1+x) = x - \frac{x^2}{2} + o(x^2), \quad \sin x = x - \frac{x^3}{6} + o(x^3), \quad \cos x = 1 - \frac{x^2}{2} + o(x^2)$$

The n-th derivative

Find the pattern from f′, f″, f‴, then prove it by induction. Useful:

$$(\sin x)^{(n)} = \sin\left(x + \frac{n\pi}{2}\right), \qquad (\cos x)^{(n)} = \cos\left(x + \frac{n\pi}{2}\right), \qquad \left(e^{ax}\right)^{(n)} = a^n e^{ax}$$

Leibniz rule:

$$(g \cdot h)^{(n)}(x) = \sum_{k=0}^{n} C_n^k \, g^{(n-k)}(x) \, h^{(k)}(x)$$

If h is a polynomial of degree d, only the terms k = 0, …, d are nonzero.

Chapter 5 — Solutions (Seminar 5)

Ex. 1

Set	Accumulation points	Why
$[0, 1) \cup \{2\}$	$[0, 1]$	each point of $[0, 1]$ is a limit of other points of $[0, 1)$; 2 is isolated
$\mathbb{Z}$	$\varnothing$	$(k - 1/2, k + 1/2)$ contains no other integer
$\mathbb{Q}$	$\mathbb{R}$	every interval contains infinitely many rationals (density)
$\{0.1, 0.11, 0.111, \ldots\}$	$\{1/9\}$	the terms increase strictly to $1/9 = 0.111\ldots$; each term is isolated

For the last set:

$$a_n = 0.\underbrace{11\ldots1}_{n} = \frac{1}{9}\left(1 - \frac{1}{10^n}\right) \longrightarrow \frac{1}{9}$$

Ex. 2

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ -1, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

- $|f| \equiv 1$, so $|f|$ is continuous everywhere.
- f is discontinuous at every x_0: take rationals $r_n \to x_0$ and irrationals $i_n \to x_0$ (density). Then $f(r_n) \to 1$ and $f(i_n) \to -1$, so the limit of f at x_0 does not exist.

Ex. 3

Let $g(x) = f(x) - x$, continuous on $[a, b]$.

$f(a) \in [a, b] \Longrightarrow g(a) = f(a) - a \geq 0$

$f(b) \in [a, b] \Longrightarrow g(b) = f(b) - b \leq 0$

$\Longrightarrow g(a) \cdot g(b) \leq 0 \Longrightarrow$ (Bolzano) there is $x^* \in [a, b]$ with $g(x^*) = 0$, *i.e.* $f(x^*) = x^*$.

Ex. 4

Continuity of f. For $x \neq 0$, f is continuous (products and compositions of continuous functions). At 0:

$$|f(x) - f(0)| = \left|x^2 \sin\frac{1}{x}\right| \leq x^2 \longrightarrow 0 \quad \Longrightarrow \quad f \text{ is continuous at } 0$$

Differentiability of f. For x ≠ 0:

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$$

At 0, from the definition:

$$f'(0) = \lim_{x \to 0} \frac{x^2 \sin \frac{1}{x} - 0}{x - 0} = \lim_{x \to 0} x \sin \frac{1}{x} = 0, \qquad \text{since } \left| x \sin \frac{1}{x} \right| \le |x|$$

So f is differentiable on ℝ.

Continuity of f′ at 0. 2x sin(1/x) → 0, but cos(1/x) has no limit at 0:

$$x_n = \frac{1}{2n\pi} \to 0 : \ \cos \frac{1}{x_n} = 1 \ \Rightarrow \ f'(x_n) \to -1; \qquad y_n = \frac{1}{(2n+1)\pi} \to 0 : \ \cos \frac{1}{y_n} = -1 \ \Rightarrow \ f'(y_n) \to 1$$

$\Longrightarrow$ the limit of f′ at 0 does not exist $\Longrightarrow$ **f′ is not continuous at 0**, so f′ is not differentiable at 0. On ℝ ∖ {0}, f′ is continuous and differentiable.

Ex. 5

The limit sin x / x. Let x ∈ (0, π/2). On the unit circle take O the origin, A(1, 0), B(cos x, sin x), and C(1, tan x) on the tangent at A. Compare the areas of triangle OAB, sector OAB and triangle OAC:

$$\frac{\sin x}{2} < \frac{x}{2} < \frac{\tan x}{2} \quad \Longrightarrow \quad \sin x < x < \frac{\sin x}{\cos x} \quad \Big| : \sin x > 0 \quad \Longrightarrow \quad \cos x < \frac{\sin x}{x} < 1$$

cos x → 1 as x → 0, because

$$0 \le 1 - \cos x = 2 \sin^2 \frac{x}{2} \le 2 \left(\frac{x}{2} \right)^2 = \frac{x^2}{2}$$

By squeeze, sin x / x → 1 as x → 0⁺. The function sin x / x is even, so the limit from the left is the same:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

(sin x)′ = cos x.

$$(\sin x)'\big|_{x=x_0} = \lim_{x \to x_0} \frac{\sin x - \sin x_0}{x - x_0} = \lim_{x \to x_0} \cos \frac{x + x_0}{2} \cdot \frac{\sin \frac{x - x_0}{2}}{\frac{x - x_0}{2}} = \cos x_0 \cdot 1 = \cos x_0$$

(cos x)′ = −sin x.

$$(\cos x)'\big|_{x=x_0} = \lim_{x \to x_0} \frac{-2 \sin \frac{x+x_0}{2} \sin \frac{x-x_0}{2}}{x - x_0} = \lim_{x \to x_0} \left(-\sin \frac{x + x_0}{2} \right) \cdot \frac{\sin \frac{x - x_0}{2}}{\frac{x - x_0}{2}} = -\sin x_0$$

Ex. 6

(a) x − 1 < ⌊x⌋ ≤ x, so for x > 0:

$$\frac{x-1}{x} < \frac{\lfloor x \rfloor}{x} \le 1 \quad \text{and} \quad \frac{x-1}{x} \to 1 \quad \overset{\text{squeeze}}{\Longrightarrow} \quad \lim_{x\to\infty} \frac{\lfloor x \rfloor}{x} = 1$$

(b) Combine the logarithms, then use the number e:

$$x\big(\ln(x+2) - \ln(x+1)\big) = x \ln\left(1 + \frac{1}{x+1}\right) = \ln\left[\left(1 + \frac{1}{x+1}\right)^{x+1}\right]^{\frac{x}{x+1}} \longrightarrow \ln e^1 = 1$$

(c) A 1^∞ limit:

$$\lim_{x\to 0} (\cos x)^{\frac{1}{x^2}} = \lim_{x\to 0} \left[\big(1 + (\cos x - 1)\big)^{\frac{1}{\cos x - 1}}\right]^{\frac{\cos x - 1}{x^2}} = e^{\lim_{x\to 0} \frac{\cos x - 1}{x^2}}$$

Use cos x = 1 − 2 sin²(x/2):

$$\frac{\cos x - 1}{x^2} = \frac{-2\sin^2 \frac{x}{2}}{x^2} = -\frac{1}{2}\left(\frac{\sin \frac{x}{2}}{\frac{x}{2}}\right)^2 \longrightarrow -\frac{1}{2} \qquad \Longrightarrow \qquad \lim_{x\to 0} (\cos x)^{\frac{1}{x^2}} = e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$$

(d) Write the power with e, then use L'Hôpital (∞/∞):

$$x^x = e^{x \ln x}, \qquad \lim_{x\to 0^+} x \ln x = \lim_{x\to 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x\to 0^+} (-x) = 0 \quad \Longrightarrow \quad \lim_{x\to 0^+} x^x = e^0 = 1$$

(e) The same idea:

$$(\sin x)^x = e^{x \ln \sin x}, \qquad x \ln \sin x = x \ln x + x \ln \frac{\sin x}{x} \longrightarrow 0 + 0 \cdot \ln 1 = 0 \quad \Longrightarrow \quad \lim_{x\to 0^+} (\sin x)^x = 1$$

(f) Let t = 1/x → 0⁺. Then x((1 + 1/x)ˣ − e) = ((1 + t)^(1/t) − e)/t. Use the Taylor expansion of ln(1 + t):

$$(1+t)^{\frac{1}{t}} = e^{\frac{\ln(1+t)}{t}} = e^{1 - \frac{t}{2} + o(t)} = e \cdot e^{-\frac{t}{2} + o(t)} = e\left(1 - \frac{t}{2} + o(t)\right)$$

$$\frac{(1+t)^{1/t} - e}{t} = \frac{-\frac{e}{2}t + o(t)}{t} \longrightarrow -\frac{e}{2} \qquad \Longrightarrow \qquad \lim_{x\to\infty} x\left(\left(1 + \frac{1}{x}\right)^x - e\right) = -\frac{e}{2}$$

Ex. 7

(a) f(x) = ln(1 + x). Compute the first derivatives:

$$f'(x) = (1+x)^{-1}, \quad f''(x) = -1 \cdot (1+x)^{-2}, \quad f'''(x) = 2\,(1+x)^{-3}, \quad f^{(4)}(x) = -6\,(1+x)^{-4}$$

Pattern:

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}, \qquad n \ge 1$$

Induction: true for n = 1. If it holds for n, differentiate once more:

$$f^{(n+1)}(x) = (-1)^{n-1}(n-1)! \cdot (-n)(1+x)^{-n-1} = \frac{(-1)^n\, n!}{(1+x)^{n+1}} \quad \checkmark$$

(b) f(x) = sin x: f′ = cos x = sin(x + π/2), f″ = −sin x = sin(x + π), …

$$(\sin x)^{(n)} = \sin\left(x + \frac{n\pi}{2}\right)$$

Induction:

$$\left(\sin\left(x + \frac{n\pi}{2}\right)\right)' = \cos\left(x + \frac{n\pi}{2}\right) = \sin\left(x + \frac{n\pi}{2} + \frac{\pi}{2}\right) = \sin\left(x + \frac{(n+1)\pi}{2}\right) \quad \checkmark$$

(c) f(x) = x² sin x. Leibniz rule with g = sin and h = x² (h′ = 2x, h″ = 2, all higher derivatives 0):

$$f^{(n)}(x) = x^2 \sin\left(x + \frac{n\pi}{2}\right) + 2nx \sin\left(x + \frac{(n-1)\pi}{2}\right) + n(n-1)\sin\left(x + \frac{(n-2)\pi}{2}\right)$$

(d) f(x) = e^(2x) x³. Leibniz rule with g = e^(2x) (its j-th derivative is 2ʲe^(2x)) and h = x³ (h′ = 3x², h″ = 6x, h‴ = 6):

$$f^{(n)}(x) = e^{2x}\left(2^n x^3 + 3n\,2^{n-1}x^2 + 3n(n-1)\,2^{n-2}x + n(n-1)(n-2)\,2^{n-3}\right)$$

Results at a glance — Ex. 6 and Ex. 7

Exercise	Method	Result
6(a) ⌊x⌋/x, x → ∞	squeeze	1
6(b) x(ln(x+2) − ln(x+1)), x → ∞	number e	1
6(c) (cos x)^(1/x²), x → 0	1^∞, number e	1/√e
6(d) x^x, x → 0⁺	e^(x ln x), L'Hôpital	1
6(e) $(\sin x)^x$, x → 0⁺	e^(x ln sin x)	1
6(f) x((1 + 1/x)^x − e), x → ∞	Taylor	−e/2
7(a) ln(1 + x)	pattern + induction	$(-1)^{n-1}(n-1)!/(1+x)^n$
7(b) sin x	pattern + induction	sin(x + nπ/2)
7(c) x² sin x	Leibniz rule	3 terms
7(d) e^(2x) x³	Leibniz rule	4 terms

Corrections made to the seminar notes

The solutions above already use the corrected versions. This table shows what changed compared with the handwritten notes.

Seminar	Exercise	In the notes	Corrected
1	1(b)	ub(B) = ℚ	ub(B) = ø
1	1(c)	lb = (−∞, −5), ub = (5, ∞)	lb = (−∞, −4], ub = [4, ∞)

Seminar	Exercise	In the notes	Corrected
1	1(d)	worked on $D = \mathbb{Q}$	the sheet has $D = \varnothing$
1	2(c)	$C = [1/2, 1)$	$C \subset [1/2, 1)$, a discrete set
1	3 in the notes (sheet Ex. 4)	$\sup(f + g) \geq \sup f + \sup g$	$\sup(f + g) \leq \sup f + \sup g$
1	interior/closure example	$\mathrm{cl}((0, 1]) = [0, 1)$	$\mathrm{cl}((0, 1]) = [0, 1]$
1	9(c)	$(-1, 1] \cap (2, \infty)$	$(-1, 1] \cup (2, \infty)$
1	9(d)	int = $\nexists$	int = $\varnothing$ (a set, not “does not exist”)
2	1	“N arbitrary”; conclusion from $\varepsilon > 0$	N chosen as $[1/\varepsilon^2] + 1$; conclusion because ε was arbitrary
2	2(a)	$x_1 > x_2 \Longrightarrow$ monotone	$x_n > x_{n+1}$ for every n
2	2(c)	$x_n = 1$, then < 1; $0 < x_n < 2$	the ratio x_{n+1}/x_n is 1, then < 1; $0 < x_n \leq 2$; lim = 0 added
2	4	$e_n \geq (1 + 1/n)^n$; $e_n = \Sigma\, 1/k!$	$e_n = (1 + 1/n)^n$; $e_n \leq \Sigma\, 1/k! < 3$
2	Stolz–Cesàro in 6(a)	denominator $z_{n+1} + z_n$	$z_{n+1} - z_n$
3	1(d)	split into divergent series	telescoping split; the sum 1/4 is unchanged
3	3(a)	$R = 1/2 \Longrightarrow$ C	$R = 1/2 < 1 \Longrightarrow$ **D**
3	3(b)	D for $a > 1/e$	D for $a \geq 1/e$
3	3(c)	$(1 + 1/(n+1))^n \to 1/e$; no conclusion	$(n/(n+1))^n \to 1/e$; C for $a < e$, D for $a \geq e$
4	2	$1/(1 - x^2)$	$1/(1 - x)^2$
4	4	Σ from n = 1 of $(-1)/(2n+1) = \pi/4$	Σ from n = 0 of $(-1)^n/(2n+1) = \pi/4$, via Abel’s theorem
4	5(b)	$x = 1/2$ gives $\Sigma\, 2^n$	$x = 1/2$ gives $\Sigma\, (-1)^n$, D
5	6(c)	exponent $(\cos x + 1)/x^2$; $\cos 2x = 1 - \sin^2 x$	$(\cos x - 1)/x^2$; $\cos 2x = 1 - 2\sin^2 x$; the result $1/\sqrt{e}$ is unchanged

Completed here, because they were unfinished in the notes: S1 Ex. 3(b) step ②; S2 Ex. 4 (the bound), Ex. 7, Ex. 8; S3 Ex. 2(b), 2(f), 3(c); S4 Ex. 5(b), 5(f); S5 Ex. 5 ($(\cos x)'$) and Ex. 7.